MIL-C-46680B(MU)
8 September 1972
SUPERSEDING
MIL-C-46680A
13 July 1964

MILITARY SPECIFICATION

CARTRIDGE, CALIBER .30, MATCH, M72

1. SCOPE

1.1 This specification covers Cartridge, Caliber .30, Match, M72 intended for competitive firing in caliber .30 match rifles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-45607	- Inspection Equipment, Acquisition Maintenance and Disposition Of

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition Through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-1168	- Lot Numbering of Ammunition

DRAWINGS

U.S. Army Munitions Command

C8595432	- Cartridge, Caliber .30, Match, M72
C7643674	- Classification of Cartridge Case Defects

FSC 1305

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2.1 (Cont'd)

F8593309 - Packing & Marking; Cartridge, Caliber .30, Match, M72; Cartons; Box, Ammunition, M2A1; Box Wirebound

F8594729 - Packing & Marking; Cartridge, Caliber .30 Match M72; 8-Round Clips (Bulk); Box, Ammunition, M19A1; Box, Wirebound

F10524016 - Packing & Marking; Cartridge, Caliber .30 Match, M72; 8-Round Clip; Box, Ammunition, M2A1; Box, Wirebound, (Bulk)

PUBLICATIONS

TECP 700-700 Vol III - Manual of Test Methods for Small Arms Ammunition

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The cartridge shall comply with Drawing C8595432, referenced specifications and the following:

3.2 Bullet extraction.-The force required to extract the bullet from the cartridge case shall be not less than 20 pounds.

3.3 Residual stress.-The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.4 Accuracy.-The average of the mean radii of all targets of the sample cartridges fired at 600 yards shall be not greater than 3.5 inches.

3.5 Velocity.-The average velocity of the sample cartridges conditioned at 70° + 2° Fahrenheit (F), shall be 2640 feet per second (fps) plus or minus 30 fps at 78 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 28 fps.

3.6 Chamber pressure.-The average chamber pressure of the sample cartridges, conditioned at 70° ± 2°F, and fired at ambient temperature, shall not exceed 50,000 pounds per square inch (psi).

3.7 Function and casualty.-The cartridge shall function without casualty.

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3.8 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.8.1 Metal defects.-The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.8.2 Foreign matter.-The cartridge shall be free of corrosion, stains, discolorations, dirt, oil, and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and consist of 500 cartridges. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval. Determination as to acceptability of any initial production sample shall be based upon results of initial tests only, and no second tests shall be permitted on that initial production sample.

4.3 Inspection provisions.

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4.3.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.2 Lot identification.-Each lot of ammunition shall be identified as to type, caliber, and model, as well as with a lot number in accordance with MIL-STD-1168. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.3 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The Acceptable Quality Level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent.

4.3.3.1 Classification of defects.-The classification of defects shall be as specified in Table I.

Table I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Cartridge				
1	Discolored, dirty, oily, smeared			X	
2	Corroded; or stained if etched		X		
	Case				
4	Round head		X		
5	Dent				X
6	Split case				
	in K, L or M location	X			
	in I, S or J location		X		
7	Perforated case	X			
8	Draw scratch				X
9	Scratch			X	
10	Beveled underside of head		X		
12	Scaly metal				X
13	No chamfer on head (rim)		X		
14	Fold			X	
15	Wrinkle			X	
16	Buckle			X	
17	Bulge			X	
18	Illegible or missing head stamp			X	

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4.3.3.1 (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
19	Defective head			X	
20	Defective mouth			X	
21	No visible evidence of mouth anneal		X		
	Bullet				
22	Dent			X	
23	Scratch			X	
24	Split bullet jacket		X		
25	Loose bullet		X		
27	Scaly metal (bullet)				X
28	Upset (crooked) point			X	
30	Blunt point			X	
	Primer				
32	No primer		X		
33	Cocked primer		X		
34	Inverted primer		X		
35	Loose primer		X		
36	Nicked or dented primer			X	
37	No waterproofing material (primer pocket joint)			X	
	Gaging				
39	Total length		X		
40	Cartridge profile failure (requiring more than 20 pounds dead weight to insert in profile and alignment gage)		X		
41	Diameter of extractor groove, max.		X		
42	Diameter of extractor groove, min.			X	
43	Diameter of head		X		
44	Thickness of head		X		
45	Length to shoulder datum		X		
46	Depth of primer		X		
	Weighing				
47	Weight, min. 2/	X			

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4.3.3.1 (Cont'd)

1/ Refer to MIL-STD-636 for visual defect standards for defects 1 through 37.

2/ Each lightweight cartridge shall be disassembled and the propellant weighed. Each such cartridge found to contain 25 grains or more of propellant shall be classed as a major defect. Any cartridge containing less than 25 grains of propellant shall be classed as a critical defect.

4.3.4 Tests.-The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.4.1 Test samples.-The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges selected shall be thoroughly mixed before being divided into samples for the various tests.

Table II

Test	Number of Cartridges	Requirement Paragraph
Bullet extraction 1/	25	3.2
Residual stress (mercurous nitrate) 1/	50	3.3
Accuracy 2/	90	3.4
Velocity 2/	20	3.5
Chamber pressure 2/	20	3.6
Function and casualty 3/ Rifle, U.S. Caliber .30, M1, National Match	240	3.7

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

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4.3.4.1 (Cont'd)

2/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first sample. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

3/ The lot shall be rejected when function and casualty defects plus firing defects observed in all other firing tests exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the first tests exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantity specified under function and casualty test, shall be fired in the weapon specified therefor. This procedure shall apply regardless of the weapon or weapons in which the firing defects occurred in the first test. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

4.3.4.2 Firing defects.-Firing defects and acceptance numbers shall be as specified in Table III.

Table III

Defects	Acceptance Number First Sample	Acceptance Number Cumulative (First & Second Sample)
1. Misfire 4/		
a. No vent hole or obstruction in vent area	0	-
b. Other	1	2
2. Bullet remaining in bore 1/	0	-
3. Primer leak		
a. Perforation in firing pin indent in primer cup	1	2
b. Escape of gas through primer cup other than 3a.	1	2
c. Escape of gas around primer cup 2/	5	9

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4.3.4.2 (Cont'd)

Defects	Acceptance Number First Sample	Acceptance Number Cumulative (First & Second Sample)
d. Blown primer	0	1
e. Primer falls out of pocket on retraction of bolt	0	1
f. Primer remains in pocket but is physically loose	1	2
4. Case casualties		
a. Longitudinal split 3/		
(1) Neck and shoulder (I&S)	9	17
(2) Body (J)	3	5
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 3/		
(1) Partial, shoulder or body (J or S)	2	4
(2) Partial body (K)	0	1
(3) Partial head (L)	0	1
(4) Complete	0	1
5. Failure to extract	0	1

1/ No second sample permitted. Lot shall be rejected.

2/ Gas escapes around more than 50 percent of periphery of cup.

3/ For location of defects indicated by letters in parentheses, see Dwg. C7643674.

4/ Each cartridge that misfires shall be disassembled and examined for presence of a vent hole in the primer pocket, or any obstruction in the vent hole area of the primer pocket that can be assignable as the cause for misfire. If the vent hole is missing or obstructed, the lot shall be rejected with no second sample permitted.

4.3.5 Packaging, packing and marking inspection.-During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test cartridge shall be classed as a critical defect. Occurrence of any type other than high pressure test shall be classed as a major defect. All nonconforming cartridges shall be rejected.

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4.3.5 (Cont'd)

Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

4.3.5 Inspection equipment.-Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.4 Test methods and procedures.

4.4.1 Bullet extraction.-The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall be not less than three nor more than six inches per minute.

4.4.2 Residual stress (Mercurous nitrate).-Tests shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.3 Accuracy.-The test shall be conducted in accordance with TECP 700-700, Vol. III utilizing a range of 300 yards or 600 yards. A range of less than 300 yards shall not be used. When the testing for accuracy is conducted on a 300 yard range, the average of the mean radii of the targets shall be multiplied by the conversion factor of 2.3.

4.4.4 Velocity.-The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.5 Chamber pressure.-The cartridges shall be drilled, using the specified equipment and the drilled hole shall be covered with a half inch square of tape. The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.6 Function and casualty.-In these firings, the weapons shall be at room temperature at the beginning of the test. Two rifles shall be used, and one-half of the quantity specified in Table II shall be fired in each. Eight cartridges (one clip) shall be fired rapid fire, and intervals between clips shall not exceed half minute. The rifles shall be cooled after every six clips.

4.4.7 Defect penalty.-In any ballistic test, except function and casualty, in which the occurrence of a firing defect prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the acceptance sample shall be penalized for such defects in accordance with Table III.

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5. PREPARATION FOR DELIVERY

5.1 Packing.-Level A - (Worldwide Shipment).-The cartridges shall be packed in accordance with Drawing F8593309, F8594729 or F10524016.

5.2 Marking and labeling.-Packing boxes shall be marked and labeled in accordance with the applicable drawing cited in 5.1.

6. NOTES

6.1 Ordering data.-Invitations for bids, requests for proposal, and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.2.1 Ammunition to be procured for Camp Perry Matches shall be packed in accordance with F10524016.

6.1.3 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provision for the submission of Acceptance Inspection Reports containing final inspection results for each lot of ammunition presented to the Government.

6.2 Head stamp.-Designation"NM" will only be applied to cartridges in lot(s) specifically destined for Camp Perry National Matches.

6.3 Hazard notice.-The cartridge described herein and certain of its components are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

Custodian:

Army - MU

Preparing Activity:

Army - MU(FA)

Project No. 1305-A717

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 65 1426 5203 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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